PHASE

1

Build the Secure World

OP-TEE OS

WHAT CHANGED

- **Target selected**
  Raspberry Pi 3, AArch64
- **Secure OS compiled**

  Cross-built on the laptop
- **Boot payload prepared**

  TF-A-compatible split images

OUTPUT TREE

generated / verified files

```
optee_os/
`-- out/arm/core/
    |-- tee.bin           [Secure World]
    |-- tee.elf           [debug image]
    |-- tee-header_v2.bin [TF-A header]
    |-- tee-pager_v2.bin  [pager payload]
    `-- tee-pageable_v2.bin

Loaded by TF-A as BL32
```

OUTCOME **Secure World binaries are available.**

BOUNDARY Compilation only; not booted.

COMPLETE

PHASE

2

Package the secure boot chain

TF-A + U-Boot

WHAT CHANGED

- **U-Boot built**
  Normal World payload: BL33
- **TF-A configured**

  OP-TEE dispatcher + BL32
- **Boot image packaged**

  BL1 and FIP combined

OUTPUT TREE

generated / verified files

```
u-boot/
|-- u-boot.bin       [BL33]
|-- u-boot             [ELF/debug]
`-- System.map        [symbols]

trusted-firmware-a/
`-- build/rpi3/debug/
    |-- bl1.bin          [early stage]
    |-- fip.bin          [BL32 + BL33]
    `-- armstub8.bin     [BL1 + FIP]
```

OUTCOME **Secure and Normal payloads are packaged.**

BOUNDARY Packaging only; not booted.

COMPLETE

PHASE

3

Prepare the Normal World kernel

Linux + standard DTBs

WHAT CHANGED

- **Kernel configured CONFIG_TEE=y, CONFIG_OPTEE=y**
- **ARM64 Linux compiled**

  Raspberry Pi 3 target
- **Existing DTS compiled**

  No new hardware or custom node

OUTPUT TREE

generated / verified files

```
Step3_Output/
|-- Image                    [ARM64 kernel]
|-- linux.config             [kernel config]
`-- dtbs/
    |-- bcm2710-rpi-3-b.dtb       [Model B]
    `-- bcm2710-rpi-3-b-plus.dtb  [Model B+]

Loaded by U-Boot with Image
```

OUTCOME **Kernel and board descriptions are verified.**

BOUNDARY Communication not yet proven.

COMPLETE

PHASE

4

Create a reproducible build base

Verify + preserve

WHAT CHANGED

- **Authoritative tree selected**
  Laptop optee_full
- **Versions compared**

  Laptop and Pi revisions match
- **Evidence preserved**

  Backups + SHA-256 checksums

OUTPUT TREE

generated / verified files

```
Step4_Verification/
|-- STEP4_COMPLETION.md
`-- verified_artifacts/
    |-- optee/
    |   `-- tee*.bin + tee.elf
    |-- tf-a/
    |   `-- armstub8.bin + fip.bin
    |-- u-boot/
    |   `-- u-boot.bin
    `-- linux/
        `-- Image + DTBs + config
```

OUTCOME **One controlled, checksummed workspace.**

BOUNDARY Quality control; no runtime feature.

COMPLETE

PHASE

5

Build the Linux user space

Buildroot runtime

WHAT CHANGED

- **Buildroot started**
  Official RPi3 target, four jobs
- **Modules generated**

  About 1,713 module-related files
- **Build stopped**

  AArch64 compiler prefix mismatch

OUTPUT TREE

generated / verified files

CURRENT (partial)

```
optee_full/
|-- module_output/  (~1,713 files)
`-- out-br/
    |-- build/
    |-- host/
    `-- target/ (incomplete)
```

EXPECTED AFTER SUCCESS

```
out-br/
|-- target/usr/sbin/tee-supplicant
|-- target/usr/lib/libteec.so
|-- target/usr/bin/xtest + examples
|-- target/lib/optee_armtz/*.ta
`-- images/rootfs.ext4
```

OUTCOME **Kernel modules and partial Buildroot tree exist.**

BOUNDARY Fix toolchain; rootfs not generated.

IN PROGRESS